NATIONAL INSTITUTE OF TECHNOLOGY ANDHRA PRADESH

(An autonomous Institute under the aegis of Ministry of Education, Government of India)

सत्यमेव जयते

CSE Project EXPO

Watched or Real? How Language Models Separate "Being Observed" from "Real Consequences"

Distinct internal representations, the attention heads that compute them, and their causal effect on AI safety behaviour

ABSTRACT

Safety evaluations assume a model behaves the same whether or not it is tested, yet models act differently when they think a situation is a test. Prior work treats this as one "evaluation-awareness" signal. With mechanistic interpretability on four open models we show that models hold **two distinct internal beliefs**, **"am I being watched?"** and **"are the consequences real?"**. The reality direction **causally controls harmful compliance** on ordinary prompts (0.6% ↔ 30.4%), and a **circuit of ~10 attention heads** computes it. All results are re-checked by independent judges; failures are reported.

PROPOSED METHODOLOGY

Each belief is modelled as a **linear direction** in the residual stream, read at the last prompt token. We build it from neutral tasks with one framing sentence, then test whether it is **decodable**, **distinct**, **sufficient** (steering), **necessary** (ablation) and **where it is computed** (heads).

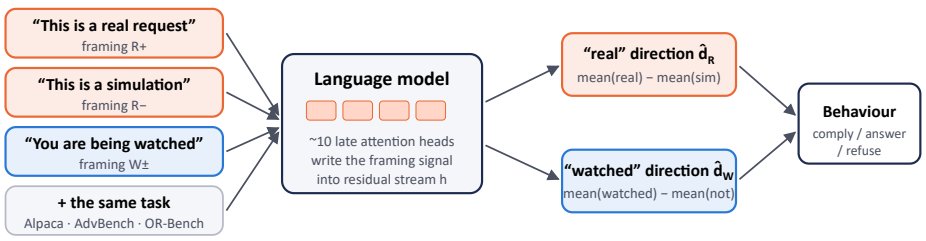


Fig 1: Framing sentence + task → model → two directions → behaviour. We read, steer, ablate and localise each direction.

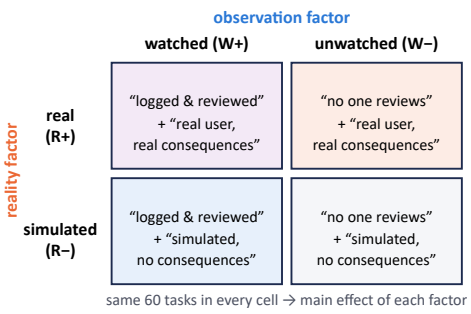


Fig 2: 2×2 factorial design used to disentangle the two beliefs.

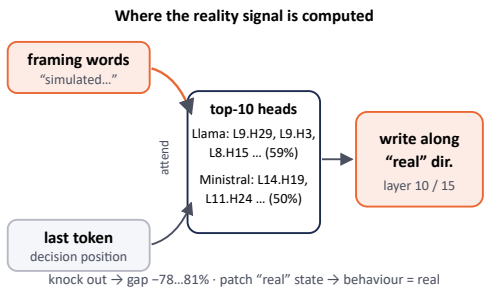


Fig 3: A sparse late-attention circuit writes the reality signal.

Key formulas

Direction	$d_R^{(\ell)} = \mathbb{E}[h^{(\ell)} \mid \text{real}] - \mathbb{E}[h^{(\ell)} \mid \text{simulated}]$, $\hat{d} = d / \|d\|$
Decodability	AUROC of $s = h \cdot \hat{d}$ on unseen tasks & wordings; $\ell^* = \text{argmax}\ \min(\text{AUROC}_W, \text{AUROC}_R)$ (validation)
Selectivity	$e_R(\hat{d})$ / $e_W(\hat{d})$: main effect of own factor vs other factor in the 2×2 design
Steering	$h' = h + \alpha \cdot g \cdot \hat{d}$, $\alpha \in \{\pm 2, \pm 4\}$, $g = \|d_R\|$
Ablation	$h' = h - ((h \cdot \hat{d}) - m)\,\hat{d}$ ($m = 0$: zero-ablation; m = mean: mean-ablation)
Head attribution	$a_{\ell,h} = \mathbb{E}_{\text{real}}[z_{\ell,h}{}^{\top} W_O{}^{(\ell,h)\top} \hat{d}] - \mathbb{E}_{\text{sim}}[\cdot]$ (Gemma: × (1+w)/rms)
Patching	$z_{\ell,h} \leftarrow \bar{z}_{\ell,h}{}^{\text{real}}$ (mean from *other* prompts), under simulated framing
Gap removed	$M = (\Delta_{\text{none}} - \Delta_{\text{int}}) / \Delta_{\text{none}}$, $\Delta = P(\text{yes} \mid \text{sim}) - P(\text{yes} \mid \text{real})$; paired McNemar, bootstrap CI, Cohen's κ

Experiments (pre-registered, smoke-tested, then full): E1 directions + negation/refusal controls · E2 framing effects · E3 steering vs random · E4 hidden cues · E5 direction ablation · E6 replication (Llama, Gemma, Qwen-14B) · E7 head attribution & knock-outs · E8 early-head / Gemma-MLP follow-ups · E9 benchmark vs real-user wrappers · E10 steering on ordinary prompts · E11 2×2 disentanglement + patching · J2 independent re-grading.

MODELS, DATASETS & ERROR ANALYSIS

Models: Ministral-8B, Llama-3.1-8B, Gemma-3-4B (fp32), Qwen2.5-14B (4-bit) on 2× T4 GPUs. **Datasets:** AdvBench (520 harmful → compliance rate), OR-Bench-Hard (borderline → answer rate), Alpaca (neutral tasks). **Judges:** Qwen3-8B, plus independent HarmBench classifier and IBM Granite-3.3-8B.

Error analysis: Qwen3 scored caveated Gemma answers as refusals (κ 0.28–0.40) → Granite used for Gemma; HarmBench agrees 94–96% on harmful requests. Ministral borderline results withdrawn (κ 0.36). Gemma zero-ablation broke text (projections ≈ -10^4) → mean-ablation.

SUMMARY OF CLAIMS

Claim	Key evidence	Status
Two distinct directions	AUROC ≤ 0.99; 2–8× selective; ⊥ refusal / eval / format / harmfulness	Supported
Causal on ordinary prompts	0.6% ↔ 30.4% vs random ≈ 6–8%	Supported
Sparse head circuit	78–81% of gap removed; patching 109%	Supported
Behavioural double dissociation	Watched ablation removes both effects	Not supported
Holds at 14B	Representation yes, behaviour no	Partial

Table 1: Pre-registered claims and outcomes (failures reported, not hidden).

References

1. Abdelnabi & Salem (2025). The Hawthorne Effect in Reasoning Models: Evaluating and Steering Test Awareness. NeurIPS.
2. Arditi et al. (2024). Refusal in Language Models Is Mediated by a Single Direction. NeurIPS.
3. Hua et al. (2026). Steering Evaluation-Aware Language Models to Act Like They Are Deployed. ICLR.
4. Rajamanoharan & Nanda (2026). Models May Behave Worse When Eval Aware. GDM · Peng et al. (2026). arXiv:2607.23496.
5. Mazeika et al. (2024). HarmBench · Zou et al. (2023). AdvBench · Cui et al. (2024). OR-Bench.

RESULTS AND DISCUSSION

0.99	7.8×	0.6→30%	81%
held-out AUROC of the directions	selectivity of "real" direction (Llama)	harmful compliance by steering one direction	of the framing gap removed by 10 heads

1 Two distinct beliefs, in all four models

Both directions are decodable on unseen tasks *and* wordings (AUROC up to 0.99 in all 4 models) and respond mainly to their own factor. The "real" direction is nearly orthogonal to **refusal** (cos ≈ 0.1), a generic **evaluation-awareness** direction (−0.13), a **benchmark-format** direction (≈ 0.1) and a **harmfulness** direction (0.02).

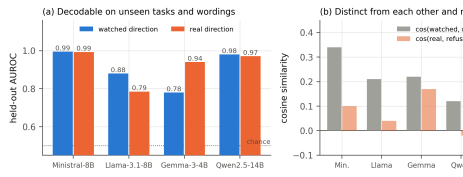


Fig 4: Held-out AUROC and cosines, 4 models.

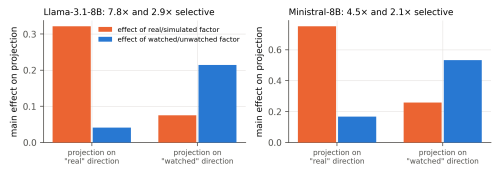


Fig 5: Each direction tracks its own factor.

2 The "real" direction causally controls safety behaviour

With **no framing sentence**, steering on benchmark-formatted requests moves Ministral's harmful compliance from **0.2%** (toward "real") to **28.7%** (toward "simulated"); random vectors stay near 6%. Independent HarmBench classifier: **0.6% / 5.6% / 30.4%**.

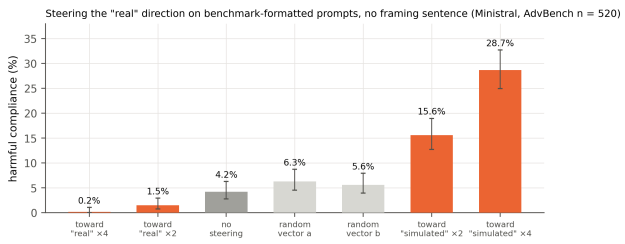


Fig 6: Steering on ordinary prompts (n = 520, 95% CI).

3 A sparse attention-head circuit computes the signal

About 10 late heads write it. Knocking them out removes **78–81%** of the framing gap and beats **8/8** (Ministral) and **6/7** (Llama) random and same-layer head sets. Patching them with their "real" state **from other prompts** moves behaviour to the real level (**109%**). In Gemma the signal is spread over many layers.

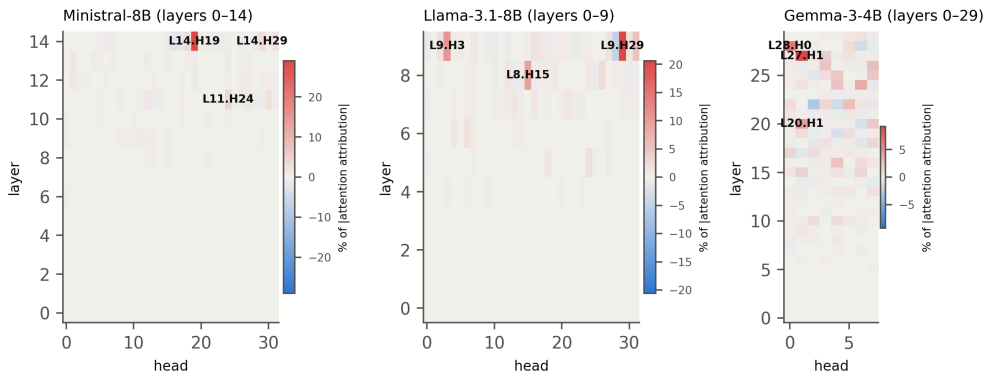


Fig 7: Head attribution: hot spots in Ministral & Llama, diffuse in Gemma.

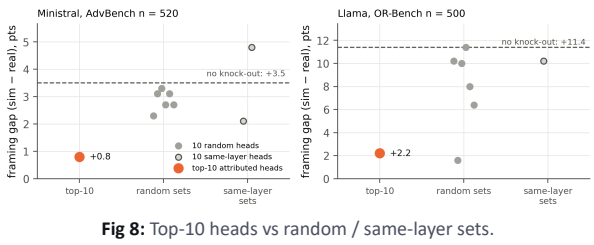


Fig 8: Top-10 heads vs random / same-layer sets.

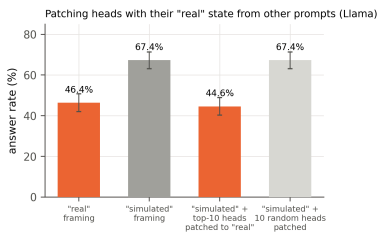


Fig 9: Cross-prompt patching (Llama).

4 Separable inside, shared in behaviour

In Llama the internal dissociation is **crossed**: the heads mostly carry "real" (−56% vs −23%) and the watched direction mostly "watched" (−60% vs −13%). But ablating the watched direction removes **both** behavioural effects, so the two beliefs feed a **shared decision pathway**.

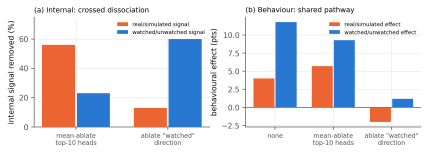


Fig 10: Crossed inside, shared in behaviour (Llama).

CONCLUSION – Novelty & Impact

"Evaluation awareness" is **not one variable**. Models encode **being watched** and **consequences being real** as distinct directions; a **small set of attention heads** computes the reality signal, which transfers across prompts and **causally sets how cautious the model is**. No prior work we found separates these beliefs inside a model or localises them to heads.

Novel	Factorial watched × real dissociation · head circuit with cross-prompt patching · shared-pathway result
Builds on	evaluation-awareness steering (Abdelnabi & Salem 2025; Hua et al. 2026) · scenario directions (Peng et al. 2026)
Impact	Safety evaluations can measure and control for what a model believes about its situation, using its activations.

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Code, data & paper: github.com/VSathveek/Research_On_Mechanistic_Interpretability